Control Estadístico de la Calidad
Examen Argumentativo

Nombre: ____ Matrícula:___________________________

Instrucciones: Muestra cómo obtienes tu respuesta. Puedes usar Excel o Minitab para tus cálculos de probabilidades o percentiles, pero no uses otros archivos o hagas uso de la IA. Respeta el código de ética del curso y del Tecnológico de Monterrey.

1. A Shewhart $\bar{X}$ chart is used with subgroup size $n = 4$. The process has $\bar{\bar{X}} = 50$ and $\bar{R} = 8.5$.

   Calculate the upper control limit (UCL) of the chart.

2. A process has specification limits LSL = 95 and USL = 125. The estimated process standard deviation is $\sigma = 2$.

   Calculate the potential capability index C_p.

3. A Shewhart control chart has center line 106, UCL = 118 and LCL = 96. A new sample statistic is 120.

   What is the best conclusion?

   (a) The process is out of statistical control.
   (b) The process is centered and capable.
   (c) The process variability has decreased.
   (d) No action is needed; the point is within limits.

4. In an inspection of 200 units, 27 defective units were found. The fraction defective is $p = 0.135$.

   Which interpretation is correct?

   (a) Exactly 86.5% of future units will be conforming.
   (b) The process is automatically in statistical control.
   (c) Approximately 13.5% of the inspected units were defective.
   (d) The process capability index equals 0.135.

5. A process has $C_p = 1.7$ and $C_{pk} = 1.4$.

   Which conclusion is most appropriate?

   (a) The process is out of statistical control.
   (b) The process spread is adequate, but the process is not perfectly centered.
   (c) The specification limits are wider than the process spread because $C_{pk} > C_p$.
   (d) The process is perfectly centered between specifications.